

AI Use Note

Q2)

i) Tool used: ChatGPT

What I asked it: Write down the rotation matrices R_{sa} and R_{sb} , is it a vector originally in s frame to be expressed in a frame or vice versa?

How I used the output (and what I changed/verified myself):
I learned the convention - its actually a vector in ' a ' frame expressed in ' s ' frame

ii) Tool used: ChatGPT

What I asked it: changing coordinates
(from the $\{s\}$ frame to $\{b\}$) without moving the point p , or as moving the location of the point without changing the reference frame of the representation? whats the diff?

How I used the output (and what I changed/verified myself):
Basically, changing coordinates is when u change the frame so the underlying basis vectors change so the coordinates change, it happens when u multiply the rotation matrix 'from' to 'to' with the vector in 'from' frame and moving the location is rotating the point in the same frame without changing the basis vector, it happens when u multiply the rotation matrix 'from' to 'to' with the vector in 'to' frame.

iii) Tool used: ChatGPT

What I asked it: changing coordinates
why rotation of point works that way when the frame remains the same but the entire point rotates how is it obtained by multiplying that matrix

How I used the output (and what I changed/verified myself):
It showed me the proof, which started with rotating a point about an axis and simplifying to get the format as in Q

Q5)

i) Tool used: ChatGPT

What I asked it: Show that $so(3)$ (the set of 3×3 skew-symmetric matrices) is a vector space. Determine its dimension and give a basis. whats a vector space? how to prove if one is a vector space

How I used the output (and what I changed/verified myself):

Its basically a collection of objects - there are 3 rules - if a subset s satisfied this then its a vector space as stated in the solution

Q7)

i) Tool used: ChatGPT

What I asked it: How to do matrix multiplication in python?

How I used the output (and what I changed/verified myself):

Its basically a collection of objects - there are 3 rules - if a subset s satisfied this then its a vector space
You do it using the '@' operator

ii) Tool used: ChatGPT

What I asked it: Pasted the py file and told 'Edit this main function inside this code so wait for 15 s first then start rotating to the 1st configuration and then one by one all in the rotation sequence array? , perform the rotation in 1 s , show the animation cleanly - just change the main function dont touch any other line

How I used the output (and what I changed/verified myself):

It generated me the main function which I pasted in the main code which worked cleanly

Q8)

i) Tool used: ChatGPT

What I asked it: Pasted the py file and Q, asked me to explain the code and what needs to be done in detail, How to do cross product and find the cap form of a vector directly in python file, also asked for the code block to save csv file.

How I used the output (and what I changed/verified myself):

I got to know cross function of numpy library and $\text{hat}(\text{vector})$ are the proper syntax - understood that we need to basically calculate the LHS and RHS of the identity and subtract them thats the residual and it should be near zero since they are equal - the csv block worked and it save the corresponding csv file

Q9)

i) Tool used: ChatGPT

What I asked it: Pasted the py file and Q, asked me to explain the code and what needs to be done in detail, How to make it repeat the sequence and how to use the set parameter option to switch between frames

How I used the output (and what I changed/verified myself):

I got to know that I need to just apply the correct matrix pre or post multiply depending on what frame is selected and the helper block code to repeat the sequence and parameter set functionality.